



Full-Stack ML System Construction

CONTENTS

Catalog

01

Introduction to Full-Stack ML

02

Phase I: Scikit-Learn & Data Pipeline

03

Phase II: Keras & Deep Learning

04

Phase III: TensorFlow & Deployment



01

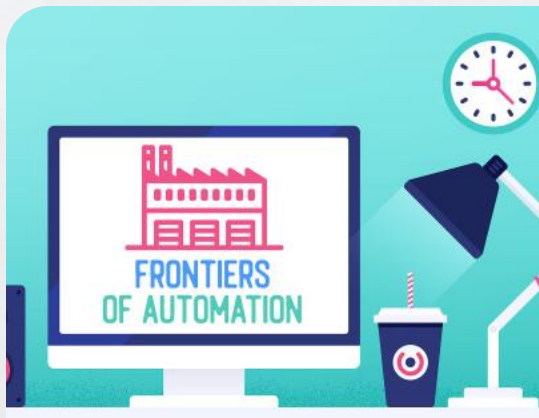
Introduction to Full-Stack ML

Goal: Production-ready AI systems



Building production-ready AI systems

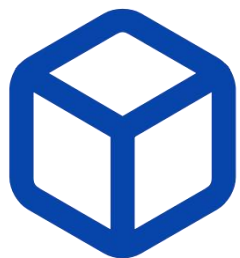
The goal is to create AI systems that are ready for real-world applications, not just academic models.



Efficient end-to-end pipelines

Implementing efficient end-to-end Machine Learning pipelines is crucial for building production-ready AI systems.

Necessity: Toolkit combination



Combining Scikit-Learn and TensorFlow/Keras

The necessity of the toolkit lies in combining Scikit-Learn for ML baseline and TensorFlow/Keras for deep learning.



Classic ML algorithms

Using classic ML algorithms like Linear Regression, SVM, Decision Trees to quickly establish baseline models.

Strategic value to saascv



Implementing efficient end-to-end pipelines

The strategic value to saascv is in implementing efficient end-to-end Machine Learning pipelines.



Combining Scikit-Learn and TensorFlow/Keras

The combination of Scikit-Learn and TensorFlow/Keras is essential for building production-ready AI systems.



02

Phase I: Scikit-Learn & Data Pipeline

Problem framing and metrics

Defining business objectives

Identifying clear business goals to guide the machine learning project

Choosing performance metrics

Selecting appropriate metrics like RMSE, Precision, and Recall for model evaluation



Data preparation core tasks



Cleaning data

Removing inconsistencies and errors to ensure data quality

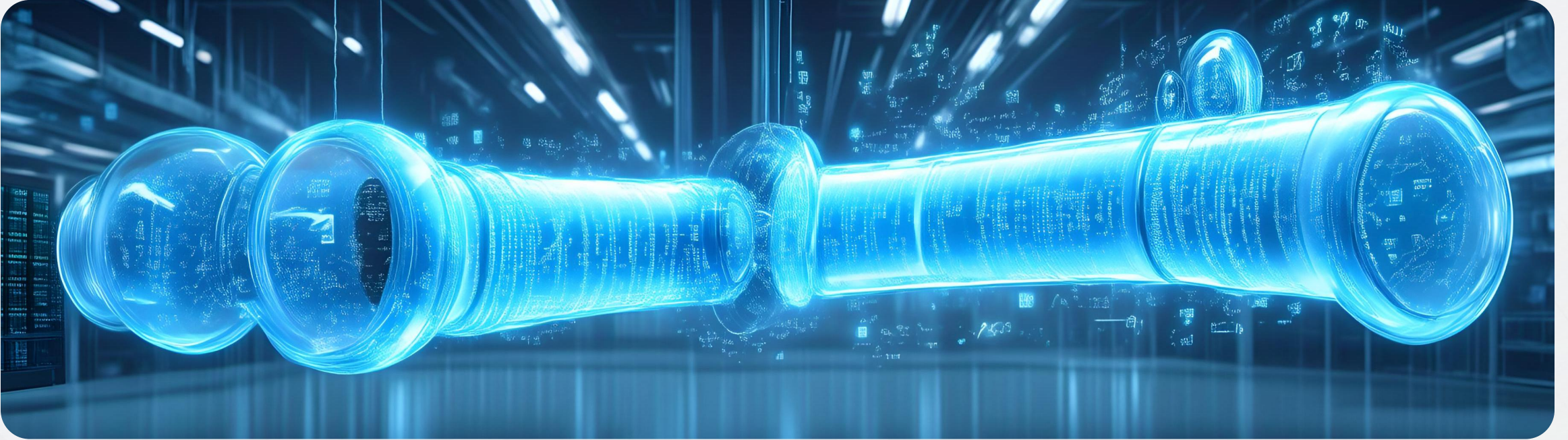
Feature scaling

Standardizing data to a uniform scale for consistent model performance

Handling categorical data

Transforming categorical variables into a format suitable for machine learning algorithms

Transformation pipeline usage



Using Scikit-Learn's Pipeline

Employing the Pipeline feature for streamlined data transformation and modeling

Baseline models establishment

Establishing performance benchmarks

Quickly setting up classic ML algorithms like Linear Regression, SVM, and Decision Trees as baseline models



03

Phase II: Keras & Deep Learning

Shift to high-dimensional data



Transition from tabular to high-dimensional

Moving from Scikit-Learn's tabular data to high-dimensional data like images/video.



Deep Learning's complexity

Addressing the complexity of high-dimensional data with deep learning techniques.

Model construction in Keras



Sequential vs. Functional APIs

Exploring the flexibility of Sequential and Functional APIs for complex architectures.



Building Blocks of Keras models

Defining layers, optimizers like Adam, and loss functions in Keras.



Integration with Scikit-Learn

Custom Keras classifiers

Creating custom classifiers to integrate deep learning models with Scikit-Learn workflow.

Workflow compatibility

Ensuring deep learning models work seamlessly within the established Scikit-Learn pipeline.

Tuning and evaluation methods



Hyperparameter Optimization

Using Grid Search and Randomized Search for hyperparameter optimization.



Evaluation techniques

Employing methods to tune and evaluate models effectively in the deep learning phase.



04

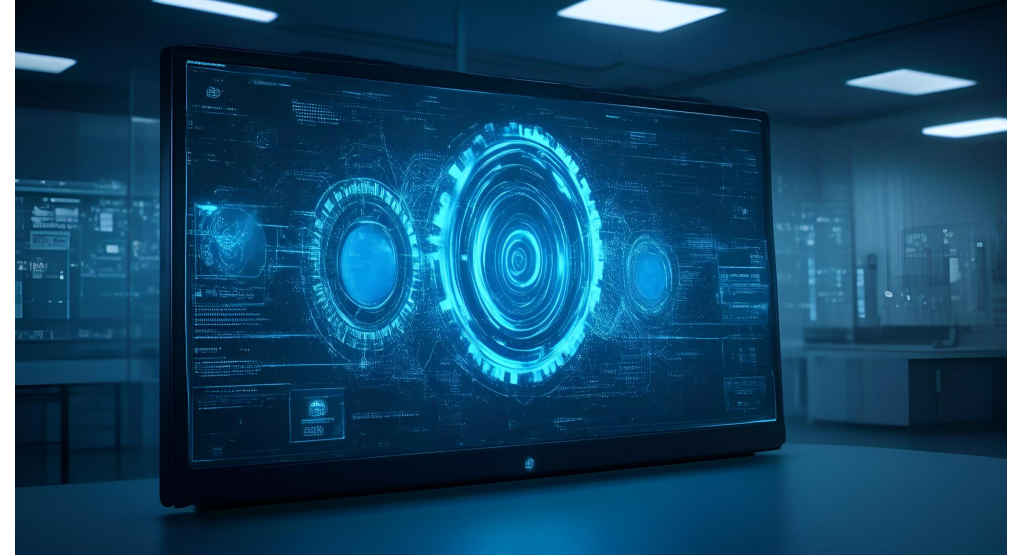
Phase III: TensorFlow & Deployment

Optimization techniques



Gradient Descent Variants

Understanding Stochastic and Mini-batch for efficient training



Efficient Training Methods

Implementing efficient training techniques to improve model performance

Model analysis tools

Callbacks and TensorBoard

Utilizing Callbacks and TensorBoard for model health monitoring

Visualization and Monitoring

Employing visualization tools to track model performance and health

Deployment toolkit options



TensorFlow Serving

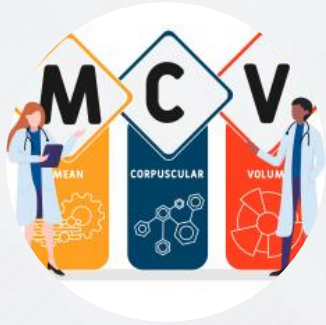
Leveraging TensorFlow Serving for scalable cloud deployment



TFLite for Edge Devices

Using TFLite for deploying models on edge devices and mobile platforms

Conclusion: Mastery essential



Three-Part System Mastery

Essential for saascv engineers to master the full-stack ML system



Production-Ready AI Systems

Ensuring the creation of production-ready AI systems, not just academic models

THE END

Thank You